Faculty of Computers and Artificial Intelligence

Image Processing (IT 443)

TA Name: ***Ahmed Ashry***

Hand Gesture Recognition

Team work:

Name	ID
Shahd Mostafa Farouk	20240806
Nourhan Noureldein Abdelhamid	20240817
Menna Talla Gamal Mahmoud	20230421
Bassant Salem Tohami	20231035
Nour Alaa Hassan	20230445

Submitted to: ***Dr. Asmaa Ahmed E. Osman***

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Hand Gesture Recognition Using PCA

American Sign Language (ASL)
Recognition Project Report

Abstract

This project presents a Hand Gesture Recognition system for American Sign Language (ASL) using Principal Component Analysis (PCA) and Deep Learning techniques. The system aims to recognize 29 ASL categories, including the English alphabet (A–Z) and three special tokens (Space, Delete, and Nothing). The dataset contains approximately 87,000 images collected from an ASL alphabet dataset in the Kaggle environment. The preprocessing pipeline includes image resizing, skin segmentation, edge detection, normalization, and augmentation to improve robustness against lighting and rotation variations. PCA is applied to reduce the dimensionality of image data while preserving 95% of the variance. In addition, a MobileNetV2 transfer learning model is used for classification. The proposed approach achieves a final training accuracy of 92.94% and a validation accuracy of 94.13%, demonstrating strong recognition performance across all 29 gesture categories.

Problem Definition and Objectives

Hand gesture recognition is an important application in computer vision and human-computer interaction. Recognizing ASL gestures automatically can help bridge communication gaps between hearing-impaired individuals and others. However, gesture recognition systems face challenges such as background noise, lighting changes, hand orientation differences, and high-dimensional image data.

The main objectives of this project are:

- To preprocess ASL images using resizing, grayscale conversion, skin segmentation, and edge detection.
- To enhance dataset diversity through augmentation techniques such as rotation, zooming, brightness adjustment, and shifting.
- To apply PCA for dimensionality reduction and extraction of the most informative features (EigenHands).
- To train a classification model capable of recognizing 29 ASL gesture categories.
- To prepare the system for future real-time webcam-based gesture recognition.

Methodology

1. Image Preprocessing

Initially, images were resized to 64×64 pixels to reduce computational complexity and improve PCA efficiency. A preprocessing pipeline was implemented using OpenCV. The pipeline applies skin segmentation in the YCrCb color space, morphological operations to clean the mask, grayscale conversion, and Canny edge detection to highlight hand contours.

2. Data Augmentation

To improve model generalization, data augmentation was performed using the Keras ImageDataGenerator. Augmentation techniques included rotation, zooming, width and height shifting, and brightness adjustment. The dataset was divided into 80% training and 20% validation samples.

3. Feature Extraction and PCA

For feature extraction, each processed image was flattened into a one-dimensional vector. StandardScaler was applied to normalize the data before PCA transformation. PCA retained 95% of the cumulative variance, significantly reducing the number of features while preserving the most important visual information. The resulting principal components are referred to as EigenHands, analogous to EigenFaces in face recognition systems.

4. Deep Learning Model (MobileNetV2)

In addition to PCA, a deep learning classification model based on MobileNetV2 was implemented using transfer learning. The pretrained ImageNet weights were used as the feature extractor, while custom dense layers, batch normalization, and dropout were added for classification. The model was trained using the Adam optimizer and categorical cross-entropy loss function. Early stopping and learning rate reduction callbacks were applied to prevent overfitting and improve training stability.

5. Real-Time Inference Pipeline

A live inference pipeline was prepared for webcam integration. A temporal smoothing mechanism based on a rolling prediction buffer was implemented to reduce prediction flickering and improve the stability of real-time gesture recognition.

Results and Evaluation

The MobileNetV2 model was trained for 10 epochs on the augmented ASL dataset. The following table summarizes the training and validation accuracy and loss at each epoch:

```
Starting Training...
Epoch 1/10
200/200 ━━━━━━━━━━━━━━━━━━━━ 264s 1s/step - accuracy: 0.6255 - loss: 1.3568 - val_accuracy: 0.8313 - val_loss: 0.7020 - learning_rate: 0.0010
Epoch 2/10
200/200 ━━━━━━━━━━━━━━━━━━━━ 234s 1s/step - accuracy: 0.8378 - loss: 0.5603 - val_accuracy: 0.8712 - val_loss: 0.4900 - learning_rate: 0.0010
Epoch 3/10
200/200 ━━━━━━━━━━━━━━━━━━━━ 224s 1s/step - accuracy: 0.8642 - loss: 0.4525 - val_accuracy: 0.7994 - val_loss: 0.6810 - learning_rate: 0.0010
Epoch 4/10
200/200 ━━━━━━━━━━━━━━━━━━━━ 223s 1s/step - accuracy: 0.8884 - loss: 0.3690 - val_accuracy: 0.8406 - val_loss: 0.5132 - learning_rate: 0.0010
Epoch 5/10
200/200 ━━━━━━━━━━━━━━━━━━━━ 221s 1s/step - accuracy: 0.8963 - loss: 0.3317 - val_accuracy: 0.9038 - val_loss: 0.3259 - learning_rate: 2.0000e-04
Epoch 6/10
200/200 ━━━━━━━━━━━━━━━━━━━━ 210s 1s/step - accuracy: 0.9091 - loss: 0.2947 - val_accuracy: 0.8938 - val_loss: 0.3526 - learning_rate: 2.0000e-04
Epoch 7/10
200/200 ━━━━━━━━━━━━━━━━━━━━ 209s 1s/step - accuracy: 0.9200 - loss: 0.2682 - val_accuracy: 0.9131 - val_loss: 0.3052 - learning_rate: 2.0000e-04
Epoch 8/10
200/200 ━━━━━━━━━━━━━━━━━━━━ 201s 1s/step - accuracy: 0.9230 - loss: 0.2635 - val_accuracy: 0.9306 - val_loss: 0.2356 - learning_rate: 2.0000e-04
Epoch 9/10
200/200 ━━━━━━━━━━━━━━━━━━━━ 197s 985ms/step - accuracy: 0.9239 - loss: 0.2483 - val_accuracy: 0.9175 - val_loss: 0.2976 - learning_rate: 2.0000e-04
Epoch 10/10
200/200 ━━━━━━━━━━━━━━━━━━━━ 199s 998ms/step - accuracy: 0.9294 - loss: 0.2332 - val_accuracy: 0.9413 - val_loss: 0.2194 - learning_rate: 2.0000e-04
```

PCA Analysis

PCA was applied to the flattened and normalized image vectors. By retaining 95% of the cumulative variance, the dimensionality of the feature space was significantly reduced from 4,096 features (64×64 pixels) to a much smaller set of principal components (EigenHands). This reduction improved computational efficiency and reduced memory requirements during training and inference without a significant loss of discriminative information.

Key Observations

1. Temporal Smoothing (Rolling Prediction Buffer)

A post-processing technique used in the real-time inference pipeline that averages predictions across a rolling window of consecutive frames, reducing flickering and improving the stability of live gesture recognition.

2. Skin Segmentation (YCrCb Color Space)

A color-based segmentation algorithm that isolates hand regions from the background using thresholding in the YCrCb color space, which is more robust to lighting variations than standard RGB.

3. Canny Edge Detection

An edge detection algorithm applied after skin segmentation to highlight hand contours and enhance feature representation for both PCA and the deep learning model.

4. Principal Component Analysis (PCA)

Dimensionality reduction technique that retains 95% of the cumulative variance, transforming flattened image vectors into compact feature representations known as "EigenHands."

5. MobileNetV2 (Transfer Learning)

A lightweight Deep Learning model that leverages pretrained ImageNet weights as a feature extractor, with custom dense layers, batch normalization, and dropout added on top for 29-class ASL gesture classification.